

Extra Regression Review

General Regression Notes

- The linear regression models we have studied estimate the mean of a quantitative response, given values of the predictors (either categorical or quantitative)
- R^2 is a measure of explained variability
- Regression can be a lot of fun!

Linear Regression with Quantitative Predictor

The first type of simple linear regression (SLR) is that with a single quantitative predictor. It always has the form

$$Y = \beta_0 + \beta_1 X$$

Where:

- Y is our outcome or dependent variable
- X is our explanatory or independent variable
- β_0 represents the y-intercept, or the mean value of Y when $X = 0$
- β_1 represents the slope. A unit change in X results in a β_1 change in Y
- R^2 gives the proportion of variability in Y that is explained by X

Example

Here we construct a model predicting penguin body mass (measured in grams) from bill length (measured in millimeters)

$$\widehat{\text{bodyMass}} = 388.8 + 86.8(\text{billLength})$$

Question 1: Using this model, interpret both the slope and the intercept. Does the intercept have a useful biological meaning? Does the coefficient for slope suggest a positive or negative relationship between bill length and body mass?

Question 2: Does have a positive coefficient guarantee that the relationship between body mass and bill length is linear? What other pieces of information do we need to confirm this?

Question 3: For two penguins whose bill lengths differ by 2mm, what difference in body mass between these penguins does the model predict?

Question 4: What is the predicted body mass of a penguin that has a bill length of 44mm? If we observe two penguins with bill lengths of 44mm, are they guaranteed to have the same body mass? Why or why not?

Question 5: What does it mean if the R^2 value for this model is $R^2 = 0.347$?

Linear regression with categorical predictor

This is just like SLR with a quantitative variable, except now our explanatory variable is categorical. It will always be of the form (e.g.,)

$$Y = \beta_0 + \beta_1 \mathbb{1}_B + \beta_2 \mathbb{1}_C$$

Where:

- Y is a quantitative outcome
- We assume the categorical variable has values A , B , and C (this will depend on the variable, this is just an example)
- β_0 contains the estimated mean for the *reference category*
- β_1 indicates the difference in mean for category B from the reference category (here, category A)
- $\mathbb{1}_B$ is an *indicator variable* for the value B
- When more than two categorical variables are used, one category from each becomes the reference

Where

$$\mathbb{1}_B = \begin{cases} 1 & \text{in category B} \\ 0 & \text{not in category B} \end{cases}$$

Remember: any value can serve as the reference variable

Example 1

First consider the values for the variables species and sex:

Table 1: Penguin Species by Sex

	Female	Male
Adelie	73	73
Chinstrap	34	34
Gentoo	58	61

Additionally, consider two separate linear models, each predicting penguin body weight in grams

Model 1:

$$\widehat{\text{bodyMass}} = 3706.2 + 26.9 \times \mathbb{1}_{\text{Chinstrap}} + 1386.3 \times \mathbb{1}_{\text{Gentoo}}$$

Model 2:

$$\widehat{\text{bodyMass}} = 3372.4 + 26.9 \times \mathbb{1}_{\text{Chinstrap}} + 1377.9 \times \mathbb{1}_{\text{Gentoo}} + 667.6 \times \mathbb{1}_{\text{Male}}$$

Question 1: What is the reference category in Model 1?

Question 2: Interpret the intercept for Model 1 and explain whether or not it has a useful biological meaning in this context.

Question 3: From Model 1, which two species appear to be the closest in average mass? How can you tell?

Question 4: Rewrite the equation for Model 1 so that Chinstrap penguins are the reference value. Does this change the predicted weight?

Question 5: What is the reference category(ies) in Model 2?

Question 6: Based on Model 2, what is the predicted difference in mass between a male chinstrap penguin and a female Gentoo penguin?

Question 7: Does Model 2 predict the same male-female difference for every species? Justify your answer.

Question 8: The R^2 value for Model 1 is $R^2 = 0.67$, while $R^2 = 0.84$ for Model 2. What does this suggest about adding sex as a variable to Model 2? Does this mean that Model 2 predicts 84% of individual penguin masses correctly?